A MULTIMODAL SYSTEM FOR AUTOMATED CORN DISEASE DETECTION AND FARMER-CENTRIC KNOWLEDGE SHARING

25-26J-239

Project Proposal Report

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Nutrient Composition Prediction and Analysis from Corn Tissue Samples

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate dissertation under my supervision.

Supervisor's Signature: ..... Date 28/8/25

Co-Supervisor's Signature: ..... Date: 29/08/25

ABSTRACT

This proposal outlines the development of a nutrient composition prediction and analysis component for corn tissue samples within a precision agriculture system. The primary purpose is to conduct a comprehensive assessment of nutrient profiles, including nitrogen, phosphorus, potassium, and zinc, using advanced machine learning techniques to predict sufficiency or deficiency based on field management data. The methodology involves detailed data collection from tissue samples at multiple growth stages, multivariate statistical analysis, and predictive modeling to identify patterns, anomalies, and inter-relationships among nutrients. This includes exploring deficiency and antagonistic effects, such as how excess phosphorus can inhibit zinc uptake. Anticipated outcomes include practical, data-informed recommendations for nutrient interventions, enabling proactive management strategies tailored to field-specific conditions like soil type, weather patterns, and farming practices. This contributes to sustainable crop nutrition by optimizing resource use, potentially reducing fertilizer application by 20-30%, minimizing environmental impact through reduced runoff, and promoting environmentally responsible farming practices that enhance long-term soil health and crop productivity.

Keywords: Nutrient prediction, Corn tissue analysis, Machine learning, Precision agriculture, Sustainable farming, Multivariate analysis, Predictive modeling

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LIST OF ABBREVIATIONS

Abbreviation	Description
N	Nitrogen
P	Phosphorus
K	Potassium
Zn	Zinc
ML	Machine Learning
NIRS	Near-Infrared Spectroscopy
Vis-NIR-SWIR	Visible-Near Infrared-Shortwave Infrared
SVM	Support Vector Machines
PLSR	Partial Least Squares Regression
CNN	Convolutional Neural Network
EONR	Economic Optimum Nitrogen Rate
RF	Random Forest
R^2	Coefficient of Determination (Model Accuracy Metric)
RMSE	Root Mean Square Error
PCA	Principal Component Analysis
R1	Reproductive Growth Stage 1 (Silking Stage in Corn)
SSNM-DSTs	Site-Specific Nutrient Management Decision Support Tools
GPU	Graphics Processing Unit
API	Application Programming Interface
KNN	K-Nearest Neighbors
IQR	Interquartile Range
DNN	Deep Neural Network

1.INTRODUCTION

1.1 Background & Literature survey

Corn (Zea mays L.) is a major staple crop globally, contributing significantly to food security, animal feed, and biofuel production, with its productivity heavily dependent on optimal nutrient management. Nutrient deficiencies, such as those in nitrogen (N), phosphorus (P), potassium (K), and zinc (Zn), can lead to reduced plant growth, lower grain quality, and substantial yield losses, sometimes up to 50% in severe cases. Traditional methods for assessing nutrient status, including soil testing and visual symptom inspections, often fail to capture real-time plant uptake dynamics, as they do not directly reflect the nutrients available to the plant during critical growth phases. In contrast, plant tissue sampling provides a more accurate and direct measure of the plant's internal nutrient status at key developmental stages, such as vegetative (V6-V9) and reproductive (R1) phases, allowing for timely corrective actions [8].

Literature extensively highlights the importance of tissue analysis for diagnosing and managing nutrient deficiencies. For instance, tissue testing offers a snapshot of nutrient content during active growth, aiding in timely interventions and preventing yield penalties [4]. Studies have shown that sampling at stages like V6 and V9 can reveal nutrient imbalances affecting biomass accumulation and grain yield, with plant analysis serving as a diagnostic tool to explain field variability in corn growth [4]. A nutrient omission experiment demonstrated how deficiencies in N, P, K, and secondary nutrients impact maize yield and nutrient balance, emphasizing the need for balanced fertilization to achieve optimal yields [6].

Recent advancements in machine learning (ML) have significantly enhanced nutrient prediction capabilities by integrating tissue data with environmental variables. ML methodologies have been applied to predict economic optimum nitrogen rates (EONR) using corn data from multiple sites, improving fertilizer efficiency and reducing over application through models like Random Forest and Neural Networks, achieving prediction accuracies with R2 values up to 0.85 [3]. Hyperspectral imaging combined with ML, such as Support Vector Machines (SVM) and Partial Least Squares Regression (PLSR), estimates nitrogen status in corn leaves with high accuracy (R2 > 0.90), using laboratory Vis-NIR-SWIR data in tropical regions [4]. Fluor sensing techniques with ML regression have been used to estimate N indicators like percentage N, above-ground biomass, and N uptake at early and mid-season stages, demonstrating robust performance across varying conditions [8]. Near-infrared spectroscopy (NIRS) has been employed for rapid protein content detection in maize grains, which correlates with nitrogen levels, achieving reliable predictions with models like Convolutional Neural Networks (CNN) [6].

Furthermore, multivariate analysis reveals complex interrelationships among nutrients, such as co-deficiencies or antagonisms (e.g., excess P inhibiting Zn uptake, or high K reducing Mg availability). A study proposed 'nutrient signatures' based on environmental impacts on N, P, K, Ca, and Mg in corn grains, using compositional data analysis to identify patterns influenced by climate and soil factors [5]. ML-based reviews on nutrient management underscore the role of data-driven techniques in estimating fertilizer needs, integrating soil, plant, and weather data for precision recommendations [1]. In hydroponics, ML optimizes nutrient uptake in crops like soybeans by predicting requirements with high precision (accuracy >95%), a concept adaptable to field-grown corn for real-time adjustments [7]. Deep learning models have been developed to determine spatiotemporal distribution of macronutrients (N, P, K, C) in cornfields using remote sensing, effectively identifying surpluses or shortages throughout growth stages [2]. Multispectral active canopy sensors combined with ML have estimated corn leaf N concentration and grain yield, with algorithms like Gradient Boosting outperforming traditional methods [8]. These studies collectively indicate a shift toward precision agriculture, where real time tissue monitoring informs fertilization, reducing waste, mitigating environmental impact such as nutrient runoff into waterways, and enhancing sustainability [5]

1.2 Research Gap

Despite significant progress in nutrient analysis and ML applications, several critical gaps remain in integrating tissue nutrient data with diverse field variables for comprehensive predictive modeling in corn cultivation. Many existing studies focus predominantly on single nutrients, such as nitrogen, while overlooking multi-nutrient interactions and their dynamics under varying farming practices, including organic vs. conventional methods, irrigation regimes, and crop rotations [1, 8]. For example, while hyperspectral and fluor sensing techniques show promise for N prediction, they often lack integration with phosphorus, potassium, and micronutrients like zinc, leading to incomplete nutrient profiles. Scaling precision tools to smallholder farms remains challenging; site-specific nutrient management decision support tools (SSNM-DSTs) struggle with adoption due to data limitations, high complexity, and insufficient validation across diverse agroecological zones, particularly in developing countries [5].

Yield gap analyses reveal substantial potential for improvement through precision methods, but current models often fail to incorporate real-time anomalies, such as those caused by extreme weather events or pest interactions and lack comprehensive frameworks linking tissue trends to long-term environmental and management data [2]. Key challenges include insufficient basic information on multi-nutrient interrelationships, in appropriate sampling protocols that do not account for spatial variability, and the absence of user-friendly, scalable ML models that provide site-specific recommendations while considering economic and environmental trade-offs. There is a pressing need for advanced ML models that predict deficiencies

proactively, incorporating anomalies, co-effects, and antagonistic interactions, to enable tailored interventions that go beyond generic advice and support sustainable practices in varied field conditions [5]. Additionally, few studies address the integration of deep learning for spatiotemporal nutrient mapping in tissue samples, limiting the ability to forecast deficiencies before visible symptoms appear.

1.3 Research Problem

The key issue is the inefficiency and limitations of traditional methods in detecting, predicting, and managing nutrient deficiencies in corn crops, which often result in suboptimal yields, excessive resource consumption, and environmental degradation. Conventional approaches like soil testing provide indirect insights and fail to capture dynamic plant uptake influenced by factors such as weather fluctuations, soil heterogeneity, and management practices, leading to over- or under-fertilization. Specifically, the research problem addresses: How can advanced computing techniques, including machine learning and multivariate analysis, be applied to tissue sample data from corn to identify intricate patterns and anomalies in nutrient levels under diverse farming practices and site conditions? Furthermore, how can these techniques formulate accurate predictive models for estimating nutrient sufficiency or deficiency based on integrated field management data, while exploring interrelationships among multiple nutrients to reveal co-deficiencies (e.g., N and S together) or antagonistic effects (e.g., high Ca suppressing Mg)? Ultimately, this problem seeks to translate analytical findings into practical, actionable recommendations for sustainable nutrient management, bridging the gap between data collection and real-world applications to enhance precision fertilization, reduce input costs by optimizing fertilizer use, and promote environmentally responsible farming that minimizes nutrient leaching and supports long-term soil fertility.

2. OBJECTIVES

2.1 Main Objectives

To develop a robust, data-driven system for predicting and analyzing nutrient composition in corn tissue samples that diagnoses the cause of deficiencies by analyzing multi-nutrient interactions, thereby enabling intelligent and sustainable nutrient management.

2.2 Specific Objectives

2.2.1 Nutrient Profile Assessment

To conduct a comprehensive assessment of nutrient profiles, including nitrogen (N), phosphorus (P), potassium (K), and zinc (Zn), across collected tissue samples. This involves laboratory analysis or spectroscopic methods to quantify nutrient concentrations at multiple growth stages and under varied field conditions.

2.2.2 Analysis of Patterns and Anomalies

To identify and interpret patterns and anomalies in nutrient levels under different farming practices, such as tillage methods and fertilizer types, and site-specific conditions like soil pH and rainfall. Statistical tools will be applied to detect deviations from optimal ranges and their potential causes.

2.2.3 Predictive Modeling for Nutrient Sufficiency

To formulate predictive approaches for estimating nutrient sufficiency or deficiency based on field management data. Machine learning models will be developed to incorporate weather, soil properties, and historical yields to forecast risks with high accuracy.

2.2.4 Exploration of Nutrient Interrelationships

To explore interrelationships among multiple nutrients to reveal co-deficiency or antagonistic effects, such as P-Zn antagonism or N-K synergies, using correlation analysis and principal component analysis (PCA).

2.2.5 Recommendations for Nutrient Interventions

To develop practical recommendations for nutrient management based on analytical findings. Guidelines will be provided for fertilizer application rates, timing, and types, considering both economic feasibility and environmental sustainability.

3. METHODOLOGY

This component will integrate seamlessly with the overall precision agriculture system by processing tissue sample data alongside a wide array of field variables (e.g., soil type, pH, moisture levels, weather data including temperature and precipitation, planting density, and historical management practices). The approach is structured in phases to ensure thorough analysis and reliable predictions, drawing on established ML pipelines.

3.1 Understanding the Research Domain

3.1.1 Data Collection

Gather tissue samples from corn plants at key growth stages (e.g., V6 for early vegetative, V9 for mid-vegetative, and R1 for silk) using standardized protocols to ensure representativeness across fields. Analyze samples for nutrients via traditional lab methods (e.g., wet chemistry for N, P, K, Zn) or nondestructive techniques like hyperspectral imaging and NIRS for rapid assessment [4, 8]. Simultaneously collect field data through IoT sensors (e.g., soil moisture probes), weather APIs, and farmer records to create a multifaceted dataset.

3.2.2 Data Preprocessing

Clean the dataset by handling missing values (e.g., imputation using mean or KNN), normalizing features to standardize scales, detecting and removing outliers via methods like Z-score or IQR, and feature engineering to derive new variables such as nutrient ratios (e.g., N:P) or vegetation indices from spectral data.

3.2.3 Data Analysis

Employ multivariate statistics, including Principal Component Analysis (PCA) to reduce dimensionality and explore nutrient interrelationships, and correlation matrices to identify co-deficiencies or antagonisms [5]. Use clustering algorithms (e.g., K-means or hierarchical clustering) to group samples by nutrient patterns, and anomaly detection techniques (e.g., Isolation Forest) to flag unusual levels linked to specific conditions.

3.2.4 Predictive Modeling

Train supervised ML models on labeled data, including regression algorithms like Random Forest for continuous nutrient level predictions, SVM for classification of deficiency states, and deep learning models (e.g., CNN for spectral data) for complex patterns [3, 6, 4]. Incorporate features from field data; evaluate models using metrics such as R2 for regression, accuracy/F1-score for classification, and RMSE for error minimization, with cross-validation (e.g., 5-fold) to ensure generalizability [8].

3.2.5 Recommendations

Develop a rule-based expert system augmented by ML outputs to generate interventions, such as “Apply 20 kg/ha N if predicted deficiency >50% and soil moisture

3.2 System Architecture Diagram

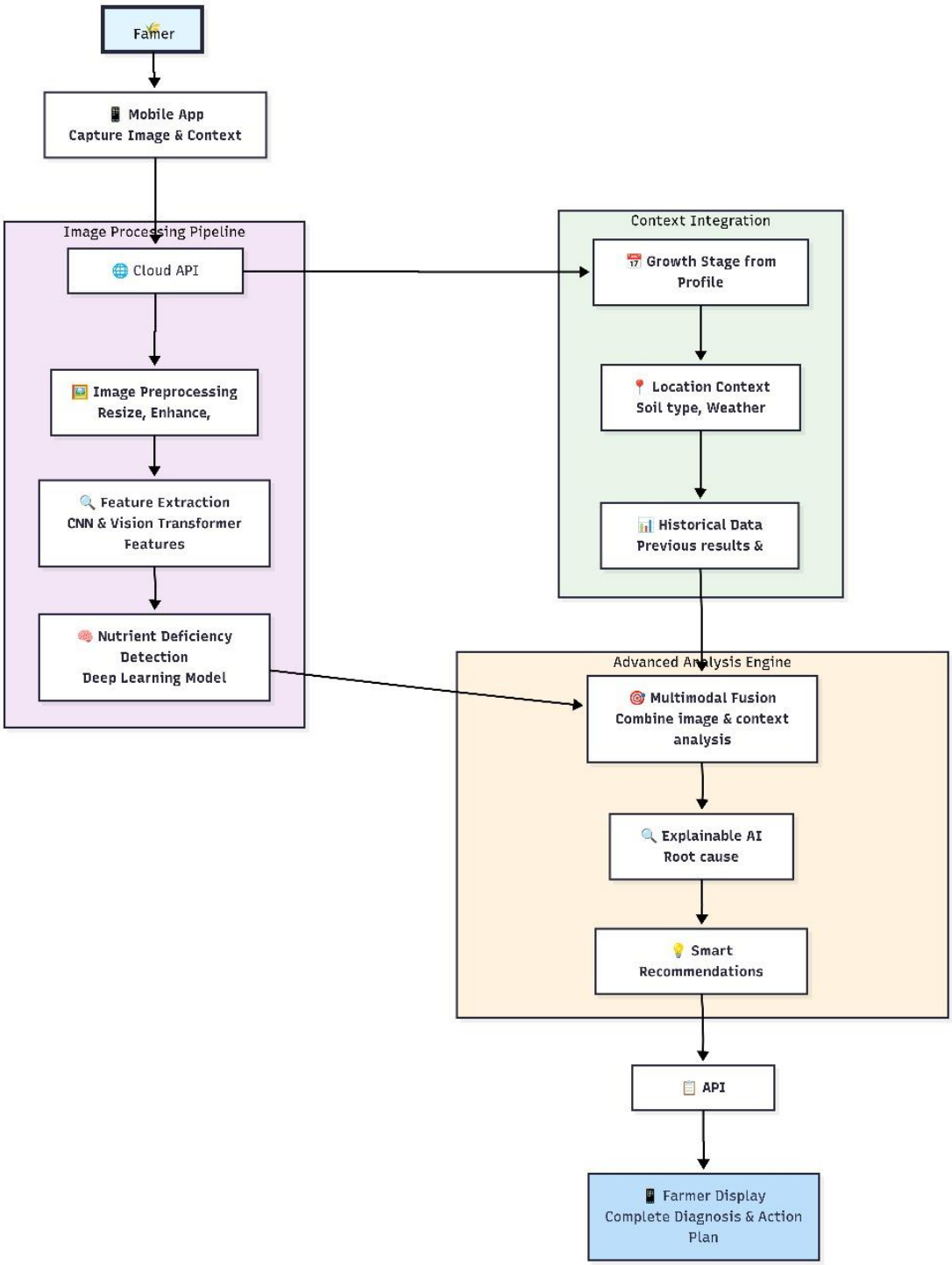

Figure 3.1: *System Diagram of Nutrient Composition Prediction and Analysis from Corn Tissue Samples*

3.3 Work Breakdown Structure (WBS)

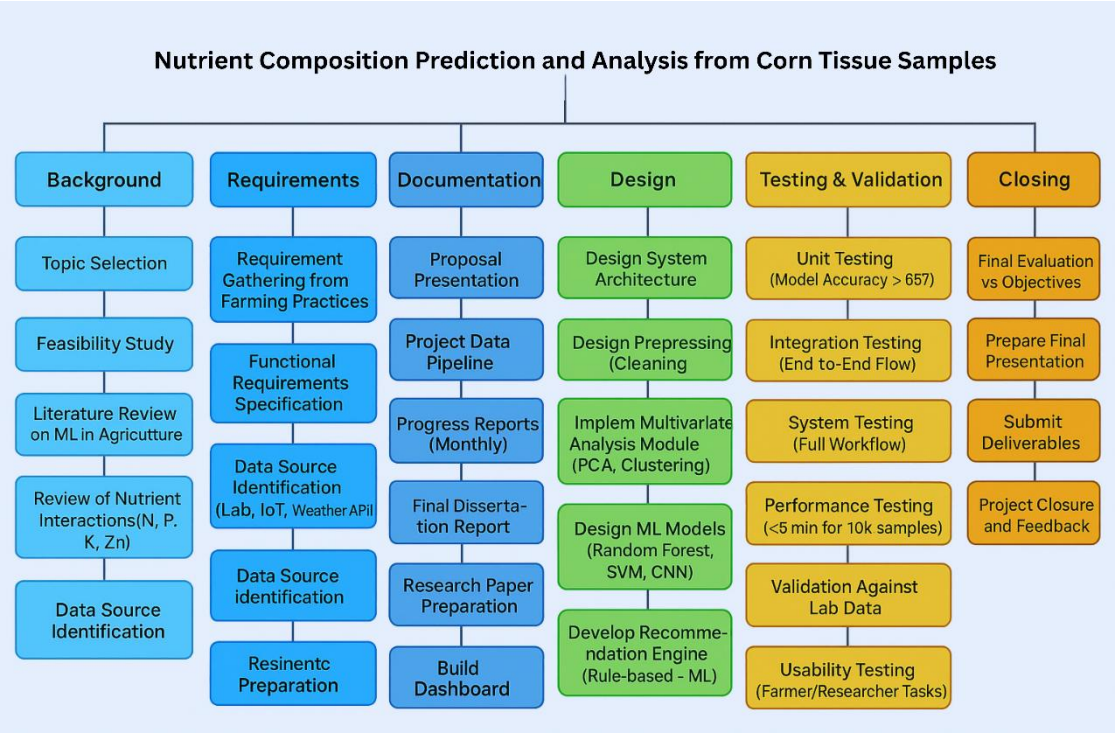

Figure 3.3: *Work Breakdown Structure (WBS)*

4.0 PROJECT REQUIREMENTS

4.1 Functional Requirements

- Image Capture & Upload
  - Support high-resolution image capture with in-app camera guidance (e.g., AR framing).
  - Allow batch upload of multiple images for efficiency.
  - Automatically tag images with GPS coordinates and timestamp.
  - Enable offline image capture with queued upload when connectivity resumes.

- AI-Powered Image Analysis
  - Integrate a trained Convolutional Neural Network (CNN) to detect nutrient deficiencies from leaf/plant images.
  - Normalize images for lighting/angle variations before analysis.
  -Output diagnosis with probability scores (e.g., "85% chance of Nitrogen deficiency").
  - Highlight affected areas on the image with visual overlays.

- Comparative & Historical Analysis
  - Compare current image analysis with past results from the same field/location.
  - Display trend graphs showing nutrient health over time.
  - Trigger alerts for deteriorating conditions or recurring issues.

- Contextual Data Integration
  - Auto-fetch local weather data (temperature, rainfall, humidity) based on GPS.
  - Integrate with soil test data (manual entry or API from labs).
  - Correlate image diagnosis with soil/weather data for higher accuracy.
  - Adjust recommendations based on crop type, growth stage, and regional norms.

- User-Centric Interface & Offline Access
  - Simple, icon-based dashboard with color-coded field health status (Red/Yellow/Green).
  - Intuitive navigation: map view → field → analysis → recommendations.
  - Full offline functionality: view past analyses, reports, and recommendations without internet.
  - Adjustable text size, voice command support, and screen-reader compatibility.

- Actionable Recommendations & Reporting
  - Generate clear, jargon-free treatment plans (e.g., "Apply 50kg/ha of Urea to Field A").
  - Recommend specific product names, brands, and application rates.

- Export/share PDF or CSV reports with images, analysis, and recommendations.
- Sync reports with Farm Management Software (FMS) via API.

- Backend & Data Management
  - Microservices architecture for scalability (auth, image processing, data fusion, reporting).
  - Hybrid database: SQL for user/field metadata, NoSQL for images and analysis results.
  - Robust sync engine for offline-to-online data reconciliation.
  - Model performance monitoring and retraining pipeline for continuous AI improvement.

- Collaboration & Community Features
  - Role-based access control for farm teams (owner, manager, contractor).
  - Assign and track field tasks linked to recommendations.
  - Securely share anonymized cases with agronomists for second opinions.
  - Optional community forum for peer knowledge sharing.

- Predictive & Proactive Features
  - Forecast disease/deficiency risk based on weather + historical trends.
  - Send early warnings (e.g., "High risk of potassium deficiency in 7–10 days").
  - Suggest preventative actions before symptoms appear.

- Integration & Extensibility
  - Open API for third-party developers and ag-tech partners.
  - Bi-directional sync with popular Farm Management Systems (e.g., John Deere Ops Center, FarmLogs).
  - Support for IoT sensor data integration (optional future phase).

4.2 Non-Functional Requirements

- Accuracy: Prediction precision >85% for key nutrients, validated against lab standards

- Efficiency: Process datasets of up to 10,000 samples in <5 minutes on standard hardware.

- Scalability: Handle increasing data volumes (e.g., 1000+ samples per session) without performance degradation

- Usability: Intuitive web-based interface with tooltips, tutorials, and mobile responsiveness.

- Security: Implement data encryption for sensitive field information and user authentication to ensure privacy.

- Reliability: System uptime >99%, with error handling for invalid inputs.
- Maintainability: Modular code structure with documentation for future updates.

5.GANTT CHART

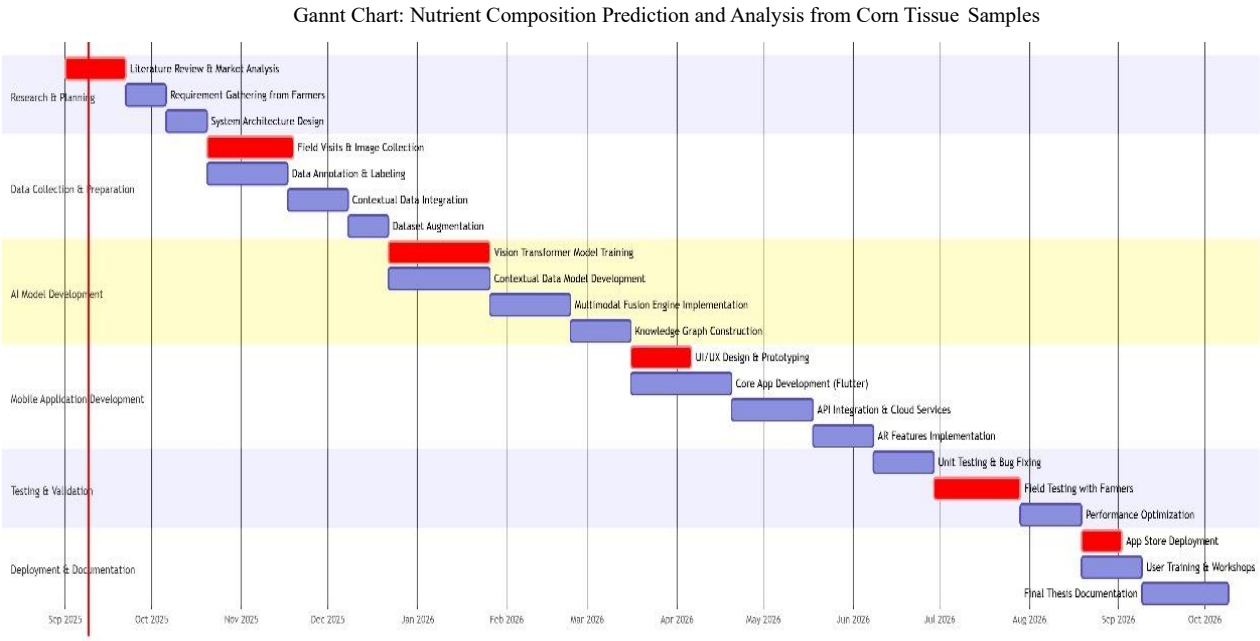

Figure 5: *Gannt Chart*

6. BUDGET AND BUDGET JUSTIFICATION

Item	Description	Quantity	Unit Cost (LKR)	Total Cost (LKR)
1. Data Collection				
1.1	Laboratory Analysis Fees (N, P, K, Zn per sample)	8 samples	2,000	16,000
1.2	Transportation for Field Visits	2 trips	2,000	4,000
			Subtotal	**20,000**
2. Software & Tools				
2.1	Open-Source Software (Python, Scikit-learn, Pandas)	1	0	0
2.2	Data Visualization (Matplotlib, Seaborn)	1	0	0
			Subtotal	**0**
3. Miscellaneous				
3.1	Printing and Binding	1 set	0	0
3.2	Contingency (0% of total)	-	-	0
			Subtotal	**0**
			GRAND TOTAL	**20,000**

- Laboratory Analysis Fees (LKR 16,000): This covers the cost of analyzing 8 corn tissue samples for the four key nutrients (N, P, K, Zn). This smaller sample size is a cost-effective approach for the initial proof-of-concept phase of the research, focusing on validating the methodology with a manageable dataset.
- Transportation (LKR 4,000): This is allocated for two essential field visits to collect tissue samples and gather management data from collaborating farms.
- Software & Tools (LKR 0): The project will utilize free, open-source software (Python and its libraries like Pandas, Scikit-learn, Matplotlib) for all data analysis, modeling, and visualization, eliminating software licensing costs.
- Printing and Binding (LKR 0): As the final report will be submitted electronically, no printing costs are incurred.

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